XINGQIANG CHEN

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OBJECTIVE

Looking for **FULL TIME** roles in **LLM Systems AI Compiler for Inference Applied Foundation Models Recommendation Systems**.

EDUCATION

PhD's in Biophysics and Condensed Matter Sep. 2014 - Jul. 2017 (Suspension)
Xiamen University, Department of Physics and Engineering GPA : 3.5/4.0

B.S. in Theoretical Physics Sep. 2010 - Jul. 2014
Xiamen University, Department of Physics and Engineering GPA : 3.0/4.0

SKILLS AND PROJECTS

Programming Languages: Proficient in Python, Java, Scala, Go.

MLOps Tools: Experienced with Hadoop, Flink, Spark, Kubernetes, KubeFlow, TensorFlow, PyTorch, and more.

LLM Inference Systems: Transformers, KV-Cache, quantization, batching & scheduling, Triton kernels, CUDA, FlashAttention, graph fusion, GPU profiling & replay.

AI Compiler Kernel: Graph compilation, persistent megakernel, operator fusion, cost modeling; LLM compiler & runtime (YiRage YPK); GPU trace capture/replay (TraceSmith).

Deep Learning Models: Knowledgeable in Transformers, RL, GNN, DNN, CNN, and more.

Interdisciplinary Expertise: Background in Physics, Computational Biology, Computational Chemistry, Molecular Dynamics, Quantum Mechanics simulations, and Computer Science.

Open Source Highlights:

- **TraceSmith** — Cross-platform GPU Profiling & Replay System for AI compilers and DL frameworks (CUDA/ROCm/Metal/M lock-free capture, deterministic replay; PyPI). `https://github.com/chenxingqiang/TraceSmith`
- **Research CLI** — Command-line AI workflow tool with academic research features; multi-model providers and MCP tooling. `https://github.com/iechor-research/research-cli`
- **JAX-sklearn** — Drop-in scikit-learn replacement with automatic JAX acceleration (*13,058 tests passed; PyPI release*). `https://github.com/chenxingqiang/jax-sklearn`
- **YouTube-DNN-RecSys** — Clean TensorFlow & Spark implementations for YouTube-style recommendation models (Python+Scala). `https://github.com/chenxingqiang/YouTube-DNN-RecSys`
- **Questions-Gen** — Math competition problem generator (Qwen3-14B, RL GRPO, KD; CLI & HF models). `https://github.com/chenxingqiang/GenQuestion`
- **DeepMicroPath** — Multi-mode research agent with microbiology specialization; non-intrusive wrapper enabling auto/default/microbiology prompts; Docker/SLURM deployment. `https://github.com/chenxingqiang/DeepMicroPath`
- **Medical-NER-Qwen-4B-Thinking** — Qwen3-4B-Thinking fine-tune via LoRA for medical entity & relation extraction (English). `https://huggingface.co/xingqiang/Medical-NER-Qwen-4B-Thinking`
- **QuestionsGen-Qwen3-14B-Final (FP)** — Full-precision merged final model trained with Unsloth; deployment-ready HF weights. `https://huggingface.co/xingqiang/QuestionsGen-Qwen3-14b-final-fp-merged`
- **YiRage Persistent Kernel (YPK)** — LLM megakernel compiler & runtime achieving 1.2×–6.7× latency reduction. `http://gitlab-repo.yizhu.local/johnson.chen/yica-yirage`

EXPERIENCE

Senior AI Compiler Engineer, Yizhu Tech Ltd. Hangzhou, China
R&D Department *Sep. 2022 - Present*

- Led development of **YiRage Persistent Kernel (YPK)** — an LLM compiler & runtime that compiles end-to-end inference into a single **megakernel**, reducing latency by **1.2×–6.7×**.
- Implemented fused operators (e.g., RMSNorm+Linear), optimized scheduling (workers & local schedulers), and persistent execution model to minimize launch overhead.
- Integrated Triton & FlashInfer kernels; added runtime profiling and cost models; enabled multi-GPU scaling and NVSHMEM-backed tensors.
- Contributed to graph compilation, kernel parameter auto-tuning, and KV-cache friendly memory layout for decoder-only LLMs.
- Published findings and filed patents on LLM inference acceleration for data center & edge deployments.
- At a ReRAM chip company, owned backend compiler adaptation and software ecosystem; focused on LLVM backend Target for the in-house SoC, driving code standardization and complex ISA development.
- Led AI model (incl. LLMs) conversion and fine-grained analysis to support multiple runtimes; evaluated operator-level metrics during inference (Cycles/MACs, etc.) and compute efficiency; integrated TVM, ONNX, MLCommons, TensorFlow/PyTorch into YZlite to collect operator metrics across runtimes and produce benchmarks on various chips.

Privacy-preserving Algorithms Engineer, Trustbe.cn
Hangzhou, China
IT Group *Sep. 2021 - Sep. 2022*

- Led research in privacy and security, focusing on homomorphic encryption, federated learning, and secure multi-party computation.
- Designed secure, privacy-preserving systems for AI applications in finance.
- Open Source Contribution: Developed **Secret-Learn**, a privacy-preserving ML library with **573 implementations** (191 algorithms × FL/SS/SL modes), 100% sklearn API compatible, JAX-accelerated (**5×+ speedup**), and integrated with SecretFlow for federated learning, secret sharing, and split learning (**PyPI release, 150,000+ lines of code**). `https://github.com/chenxingqiang/Secret-Learn`
- Published research outcomes and patented innovative work.

Senior Algorithms Engineer, Yalla Group (NYSE: YALA)
Hangzhou, China
Data Center Engineering Group *Sep. 2020 - Sep. 2021*

- Developed and improved the ChatRooms Recommendation System for Yalla App, enhancing user retention and activation time.
- Created a User Risk Control System for Ludo App, reducing users' refund rate.
- Built a GAME AI Robot for card games, improving user engagement.

Senior Algorithms Engineer, Gejia Network Tech
Hangzhou, China
Searching&Recommendation Engineering Group *Mar. 2019 - Sep. 2020*

- Developed recommendation modules for e-commerce scenarios, enhancing CTR and CVR.
- Improved recommendation system performance through multi-objective models and strategy optimization.

Machine Learning Engineer, Bianlifeng Tech
Beijing, China
Data Engineering Group *Mar. 2017 - Mar. 2019*

- Managed the Sales Forecasting Project, achieving precise demand and supply outcomes for 50 stores.
- Established the Customers&Products Recognition Project, linking customer IDs with purchases.
- Developed a model to detect and recognize goods, assisting in express checkout.

UNDERGRADUATE PROJECT

Molecular Dynamics Simulation of Proteins with QM/MM model
Xiamen university, China
(China National Funds) *Sep. 2014 - Mar. 2017*

- Conducted research on protein dynamics simulations using quantum mechanics, and analyzed biological information of proteins. This involved interdisciplinary knowledge in theoretical physics, structural biology, computational chemistry, and computer science.

Development of Liquid Biopsy ct-DNA Kit and DNA Methylation Data Analysis
Lepu Medical, China
(R&D Engineer Intern) *Mar. 2016 - Mar. 2017*

- Worked on the development of liquid biopsy kits, which required specific biomarkers for cancer screening. Using existing gene chip data (TCGA&GEO), we analyzed 10,000 samples to locate the methylation site on the Septin9 human gene, an early indicator of colon cancer. This informed the development of a more accurate clinical diagnostic kit for colon cancer.

AWARDS

- Judge, First Screening on Best Practices of Applied Algorithms, 2021.
- Ranks 57 and 32 out of 1739, IEEE ICDM Global Meteorological AI Challenge, 2018.
- Meritorious Winner, Mathematical Contest in Modeling, North American College Students, 2013.
- 6th place, International Solar Energy Decathlon Competition, 2013.
- Outstanding Prize, Contemporary Undergraduate Mathematical Contest in Modeling, 2012.

PUBLICATIONS

- Cai Y, Su M, Ahmad A, Hu X, Sang J, Kong L, Chen X, Wang C, Shuai J, Han A. Conformational dynamics of the essential sensor histidine kinase WalK. *Acta Crystallographica Section D: Structural Biology*. 2017 Oct 1;73(10):793-803.
- Xia H, Chen X, Chen X, Cheng J, Liu Y, Chen X, Zhang Q, Zou G. Chiral discrimination of amino acid enantiomers based on different interactions with Cu2+. *Sensors and Actuators B: Chemical*. 2018 Jan 1;254:44-51.
- Ahmad A, Cai Y, Chen X, Shuai J, Han A. Conformational dynamics of response regulator RegX3 from Mycobacterium tuberculosis. *PloS one*. 2015 Jul 22;10(7):e0133389.
- Chen Y, Qi H, Li X, Cai M, Chen X, Liu W, Shuai J. Suppressing effect of C a 2+ blips on puff amplitudes by inhibiting channels to prevent recovery. *Physical Review E*. 2016 Aug 15;94(2):022411.